The Extremal Principle — problem 3.32

A. Engel — Problem-Solving Strategies · 1998

Find all positive solutions of the system of equations

$$x_1 + x_2 = x_3^2, \quad x_2 + x_3 = x_4^2, \quad x_3 + x_4 = x_5^2, \quad x_4 + x_5 = x_1^2, \quad x_5 + x_1 = x_2^2.$$